TRACING

1. ADDITION OF 2 UNARY NUMBERS

1 1 1 + 1 1

Step	State	Tape	Head
0	q_0	$\underline{1}11 + 11$	0
1	q_0	$1\underline{1}1 + 11$	1
2	q_0	$11\underline{1} + 11$	2
3	q_0	$111\ \underline{+}\ 11$	3
4	q_1	$111\ \underline{1}\ 1$	4
5	q_1	$1111\underline{1}$	5
6	q_1	$111\ 11\ \underline{\ }$	6
7	q_2	$1111\underline{1}$	5
8	q_{accept}	$1111\ \underline{\ }$	6

2a. Palindrome (ACCEPT case) : abcba

Step	State	Tape	Head	NOTE
0	q_1	_bcba	1	read a, erase, move to q_1
1	q_1	_bcba	2	Scan right
2	q_1	_bcba	3	scan right
3	q_1	_bcba	4	scan right
4	q_1	_bcba	5	scan right, hit blank next
5	q_2	_bcba	4	step back to last a
6	q_3	_bcb_	3	sweep left
7	q_3	_bcb_	2	Sweep left
8	q_3	_bcb_	1	sweep left, hit blank next
9	q_3	_bcb_	0	back at start, read b
10	q_4	__cb_	1	erase b, move to q_4
11	q_4	__cb_	2	Scan right
12	q_4	__cb_	3	Scan right, hit blank next
13	q_4	__cb_	4	step back to last b
14	q_5	__cb_	3	step back to last b
15	q_3	___c__	2	matched b, erase it
16	q_3	___c__	1	sweep left, hit blank next
17	q_0	___c__	2	back at start, read c
18	q_6	_____	3	centre marker reached, erase it
19	–	_____	–	blank next → ACCEPT

2b] Palindrome Checker – Reject Case

Step	State	Tape	Head
0-6	$q_0 \to q_1 \to q_2 \to q_3$	outer a matches, erased : _bca_	3
7-11	$q_3 \to q_0$	sweeps back, sees b, goes q_4	2
12-14	q_4	scans right to blank	3
15	q_5	expects b, reads a → mismatch → REJECT	4